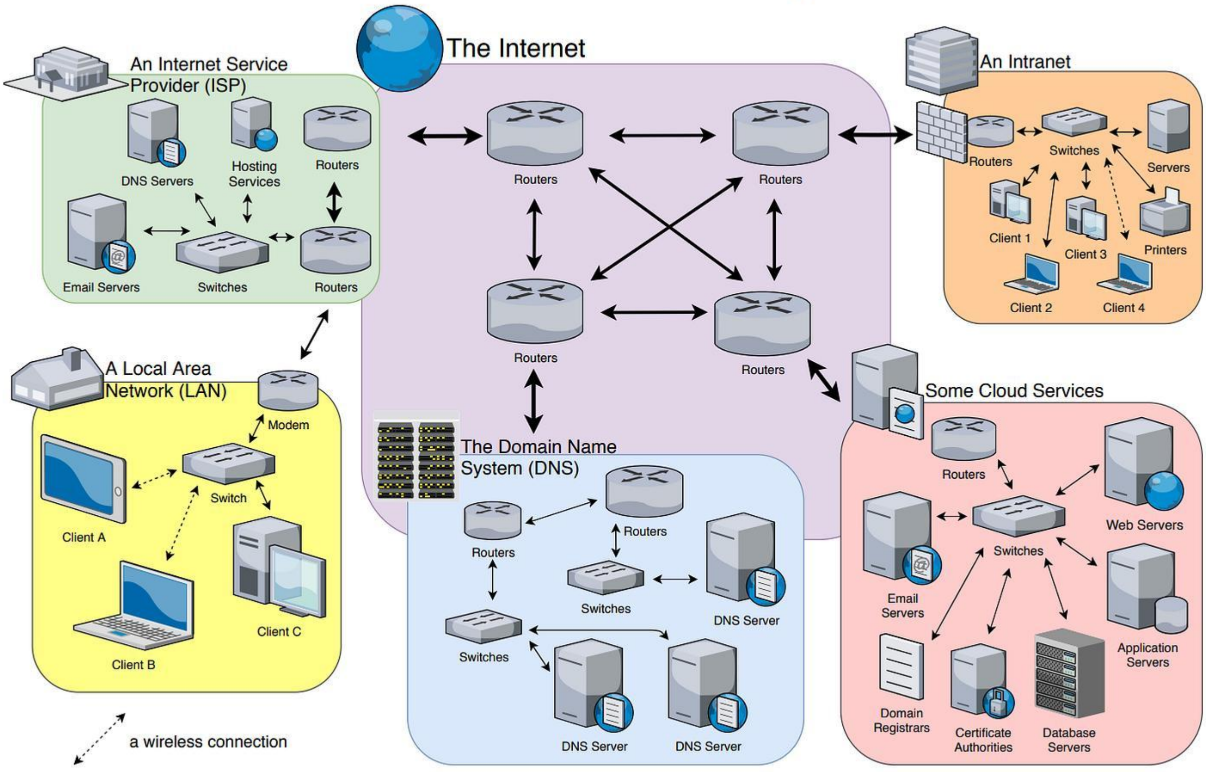
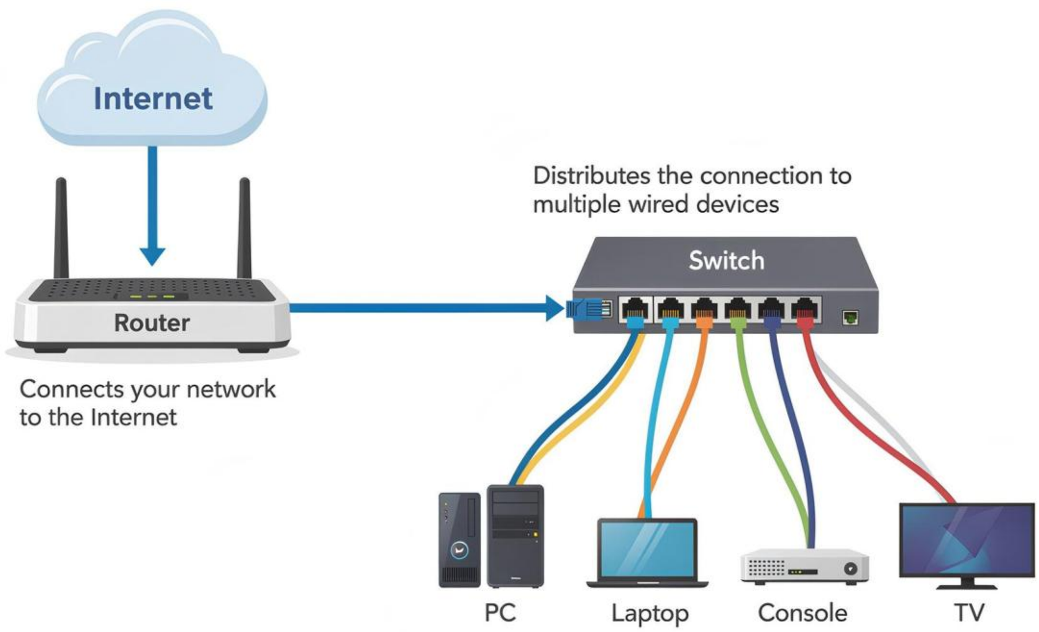
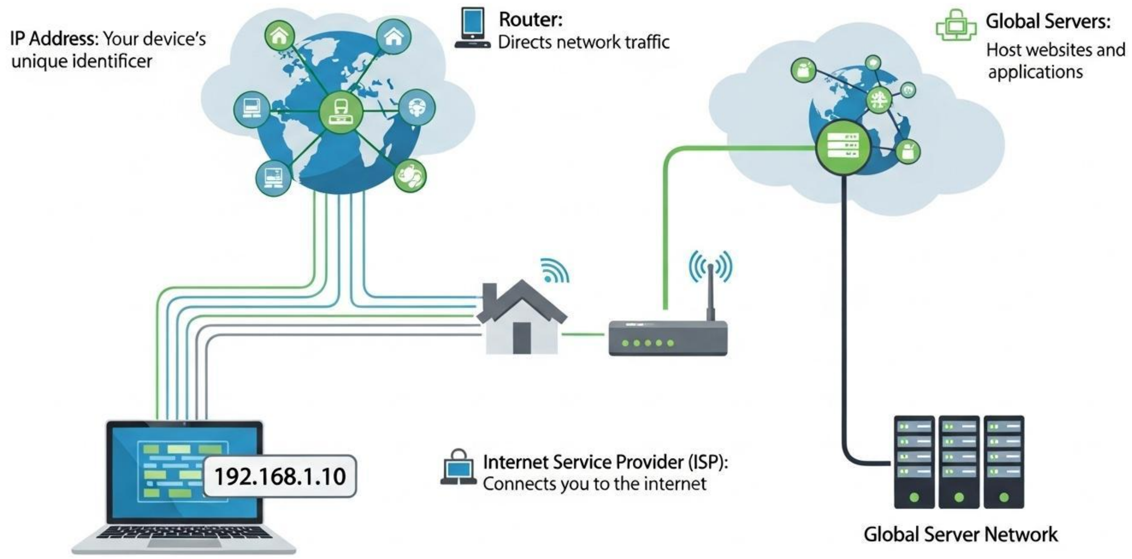
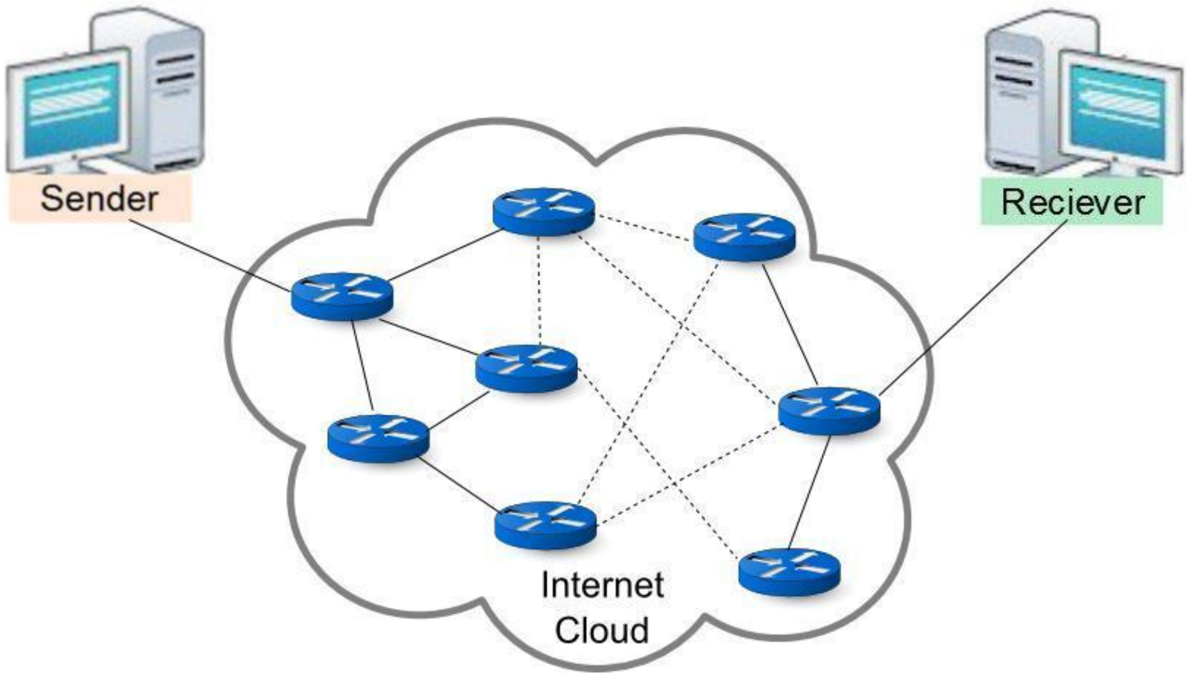
The Internet Infrastructure: *A bird's eye view*

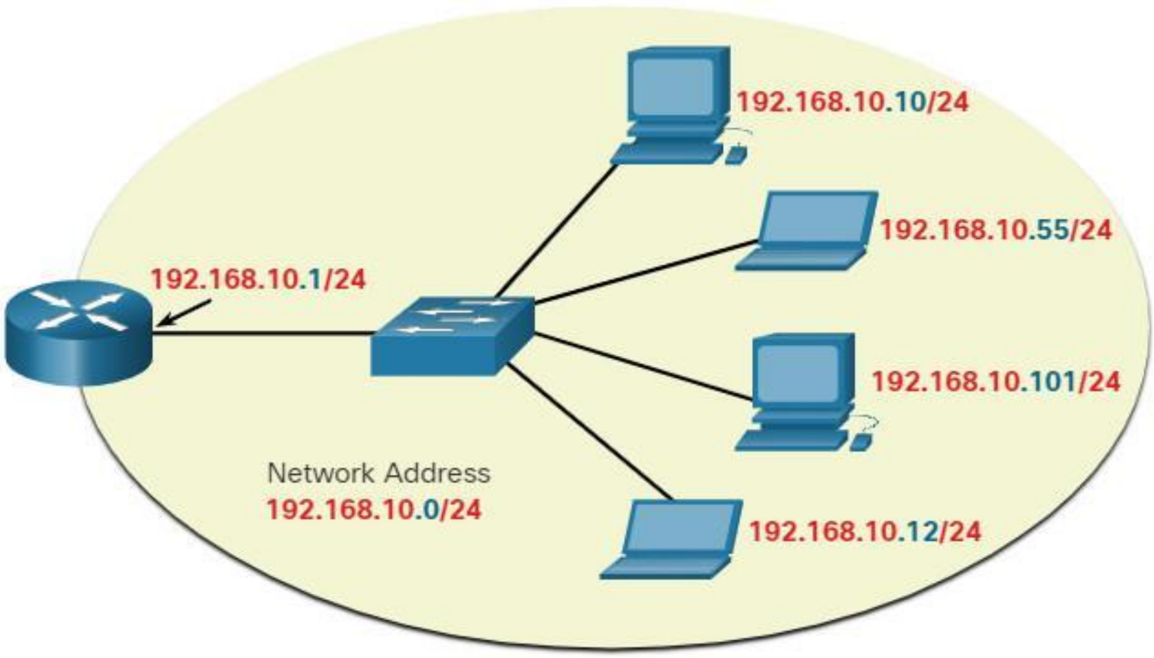


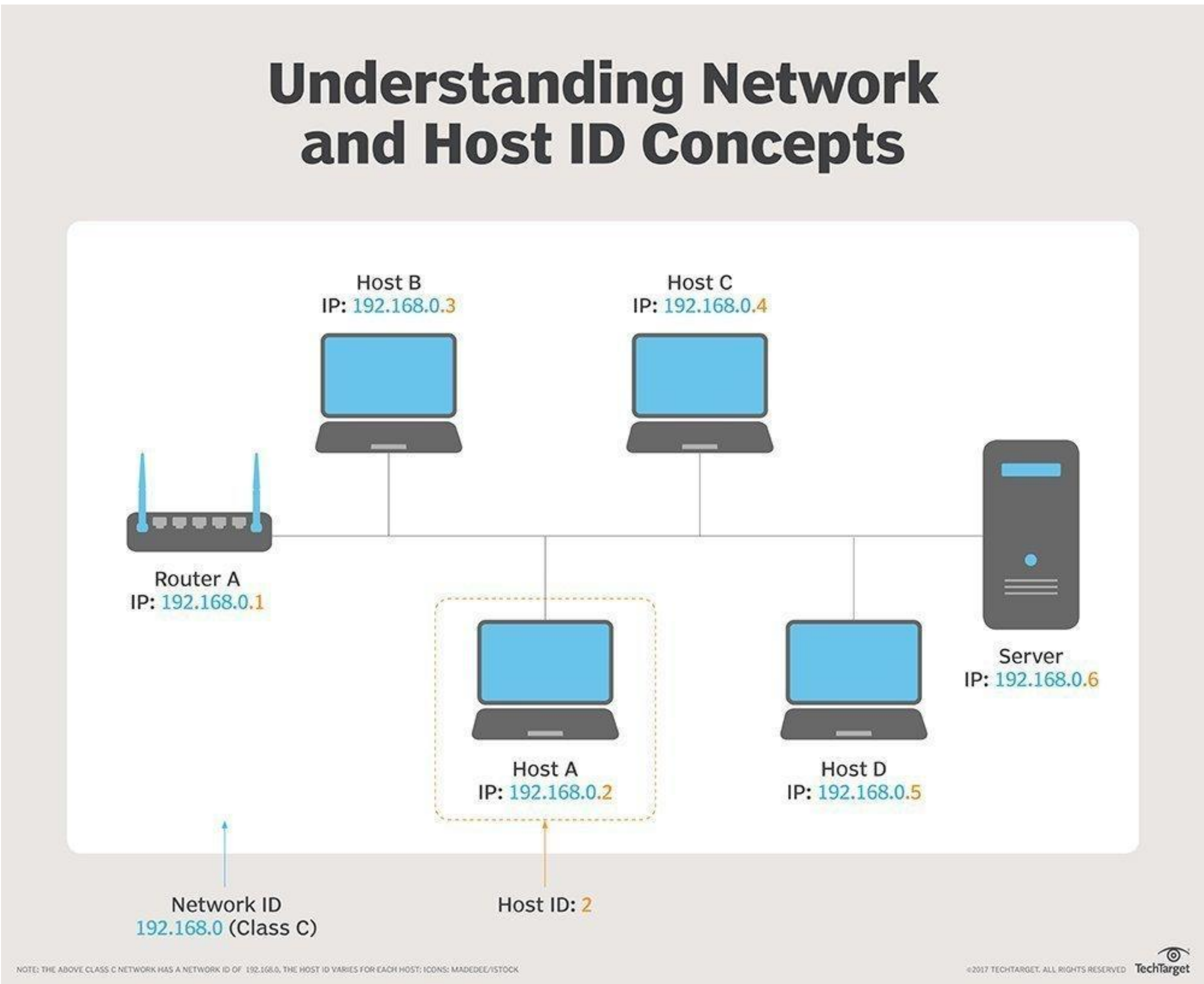
Source: [Vahid Dejwakh](), 2020




IP Address: Your device's unique identificer
Router: Directs network traffic
Global Servers: Host websites and applications
192.168.1.10
Internet Service Provider (ISP): Connects you to the internet
Global Server Network


Sender
Reciever
Internet Cloud


192.168.10.10/24
192.168.10.55/24
192.168.10.1/24
192.168.10.101/24
Network Address
192.168.10.0/24
192.168.10.12/24


Understanding Network and Host ID Concepts
Host B
IP: 192.168.0.3
Host C
IP: 192.168.0.4
Router A
IP: 192.168.0.1
Server
IP: 192.168.0.6
Host A
IP: 192.168.0.2
Host D
IP: 192.168.0.5
Network ID
192.168.0 (Class C)
Host ID: 2
NOTE: THE ABOVE CLASS C NETWORK HAS A NETWORK ID OF 192.168.0. THE HOST ID VARIES FOR EACH HOST; ICONS: MADEDEE/ISTOCK
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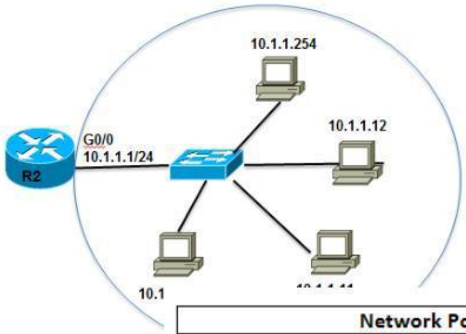
TechTarget

IPv4 Subnet Mask

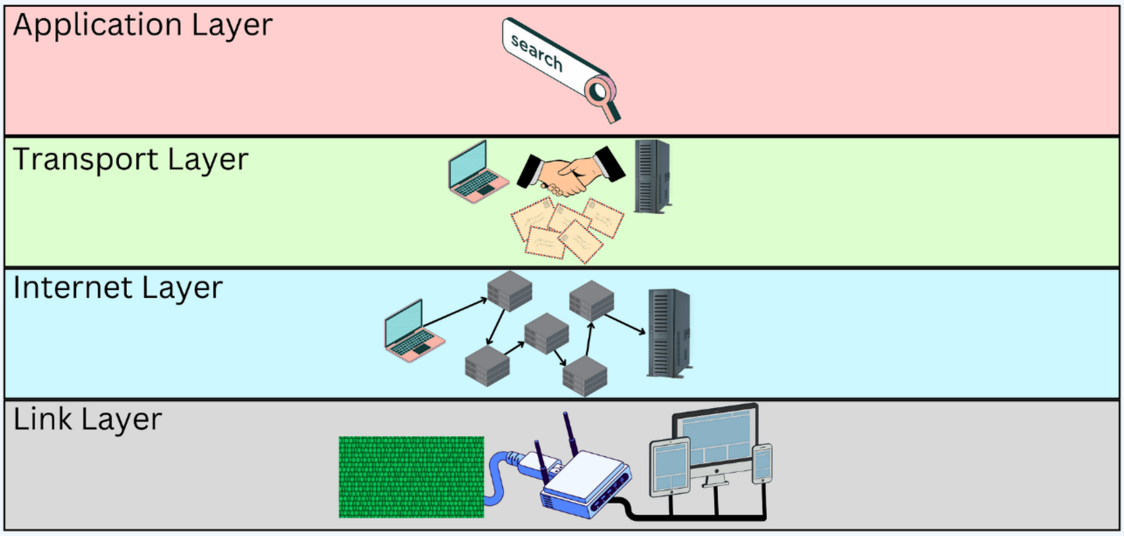
First Host and Last Host Addresses

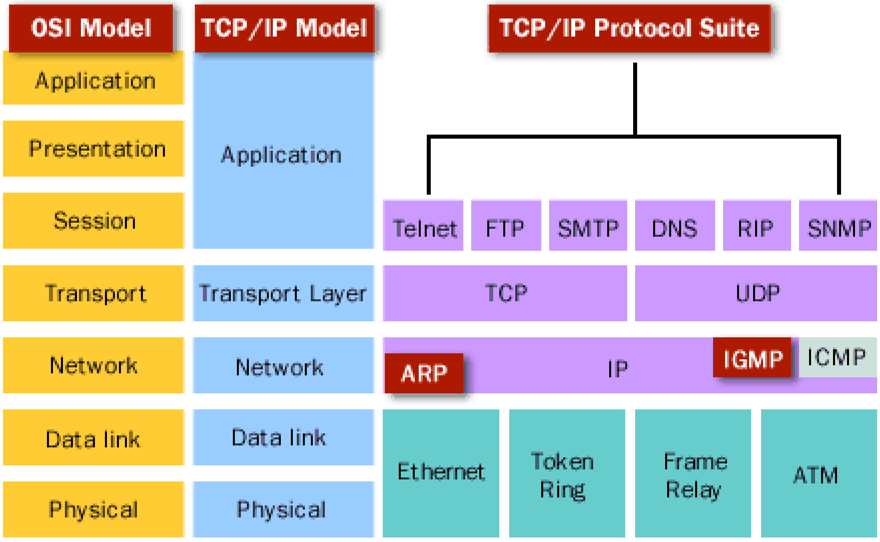
10.1.1.0/24 Network

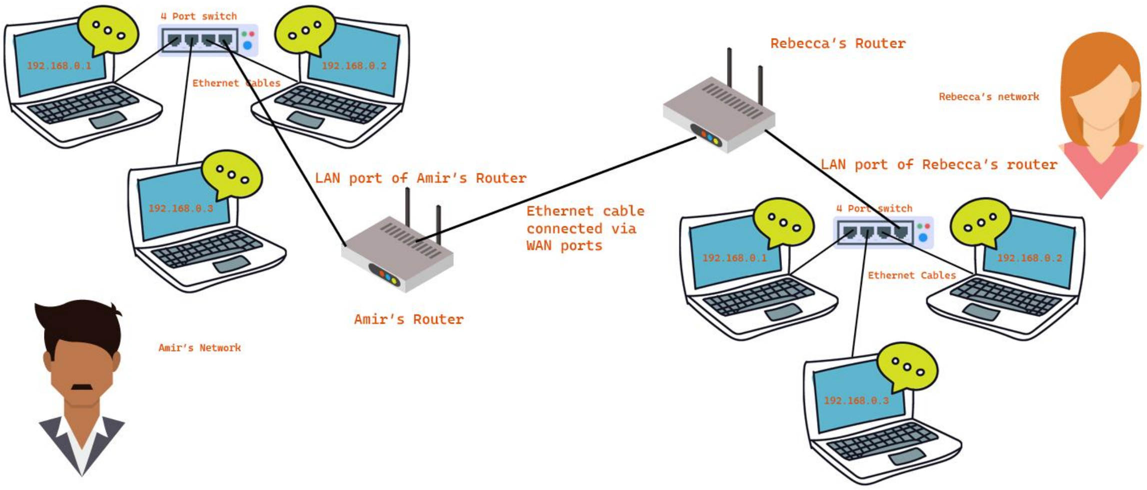


Network Portion			Host Portion	
10	1	1	1	FIRST HOST
00001010	00000001	00000001	00000001	All 0s and a 1 in the host portion
10	1	1	254	LAST HOST
00001010	00000001	00000001	11111110	All 1s and a 0 in the host portion

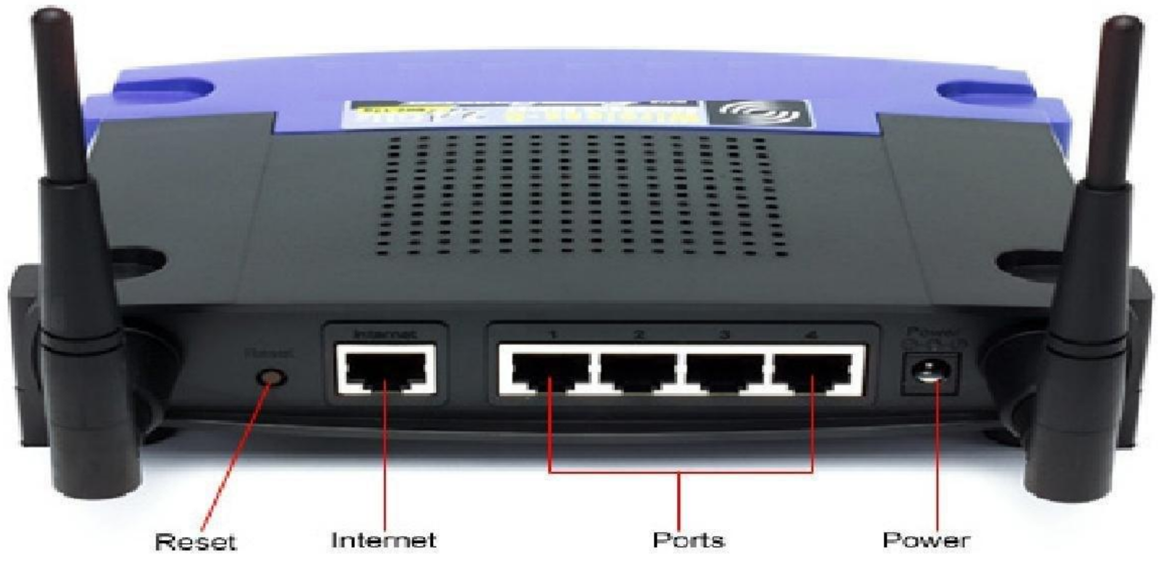
Internet Protocol Suite

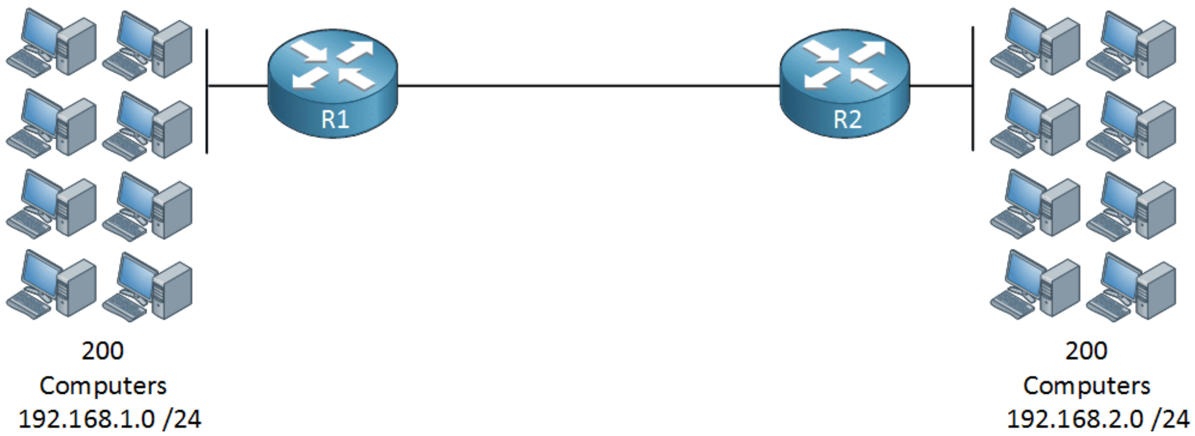


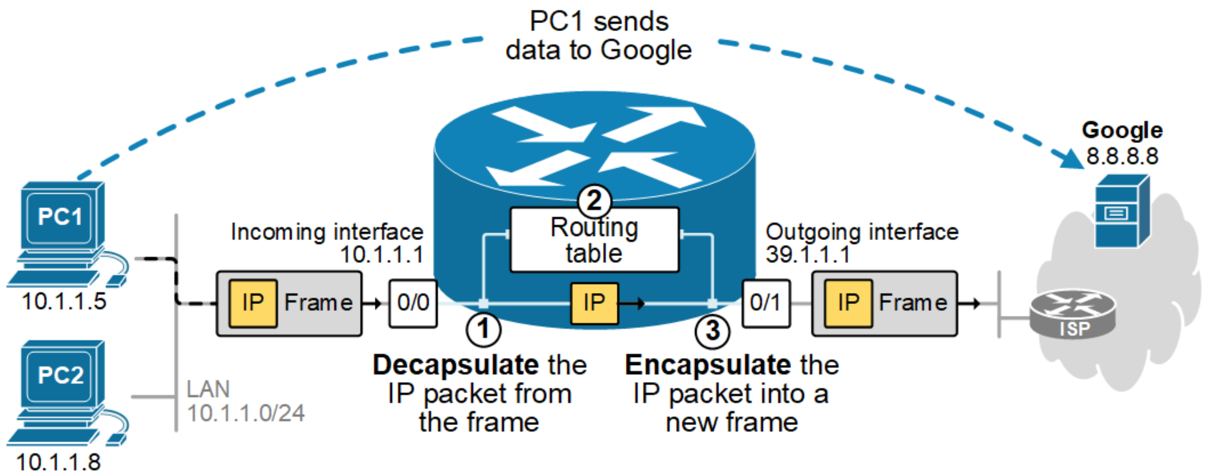

OSI Model
Application
Presentation
Session
Transport
Network
Data link
Physical
TCP/IP Model
Application
Transport Layer
Network
Data link
Physical
TCP/IP Protocol Suite
Telnet
FTP
SMTP
DNS
RIP
SNMP
TCP
UDP
ARP
IP
IGMP
ICMP
Ethernet
Token Ring
Frame Relay
ATM

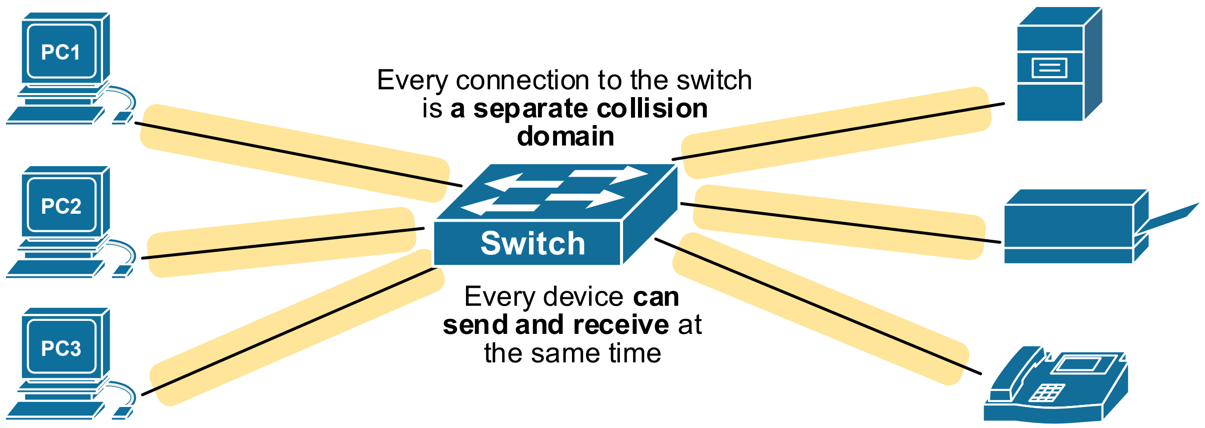

4 Port switch
192.168.0.1
192.168.0.2
192.168.0.3
Ethernet Cables
LAN port of Amir's Router
Amir's Router
Amir's Network
Ethernet cable
connected via
WAN ports
Rebecca's Router
Rebecca's network
LAN port of Rebecca's router
4 Port switch
Ethernet Cables
192.168.0.1
192.168.0.2
192.168.0.3

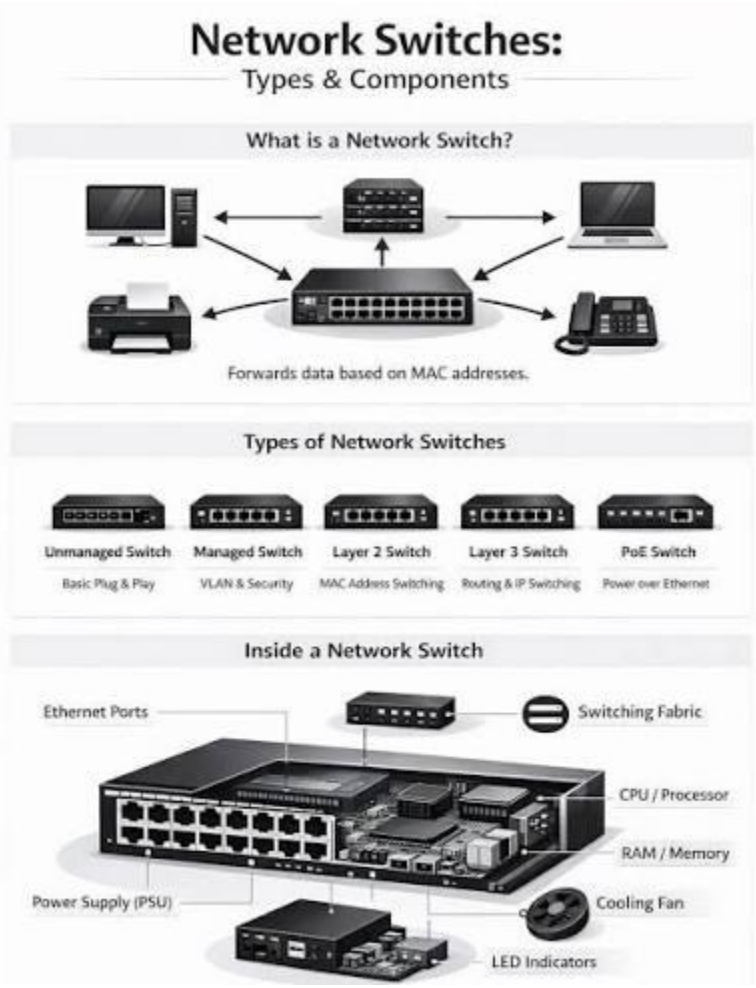

AC1200

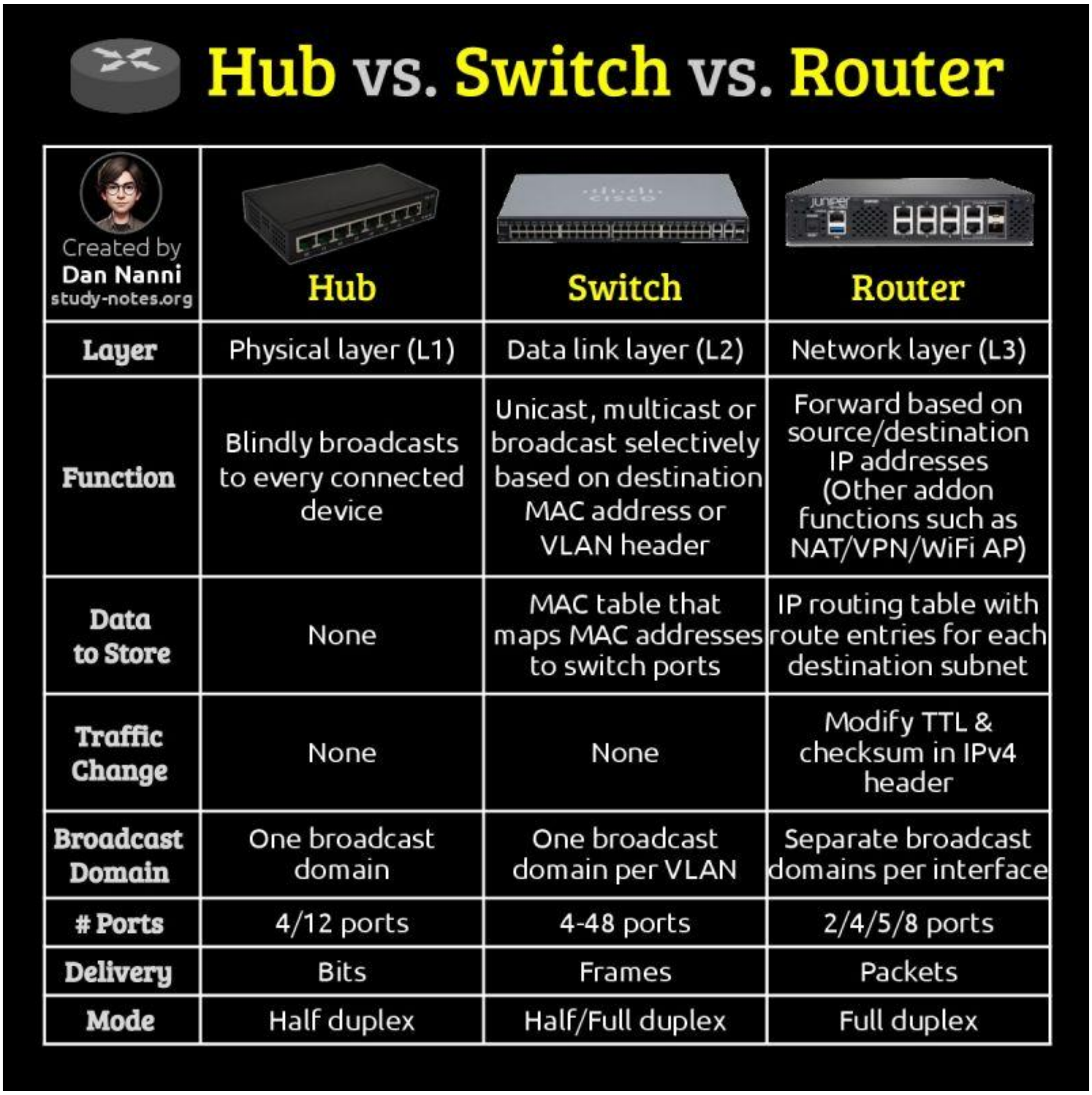

Internet
1
2
3
4
Reset
Internet
Ports
Power


R1
R2
200
Computers
192.168.1.0 /24
200
Computers
192.168.2.0 /24


PC1 sends
data to Google
Google
8.8.8.8
PC1
10.1.1.5
PC2
10.1.1.8
LAN
10.1.1.0/24
Incoming interface
10.1.1.1
IP
Frame
0/0
2
Routing
table
IP
0/1
Outgoing interface
39.1.1.1
IP
Frame
ISP
1
Decapsulate the
IP packet from
the frame
3
Encapsulate the
IP packet into a
new frame

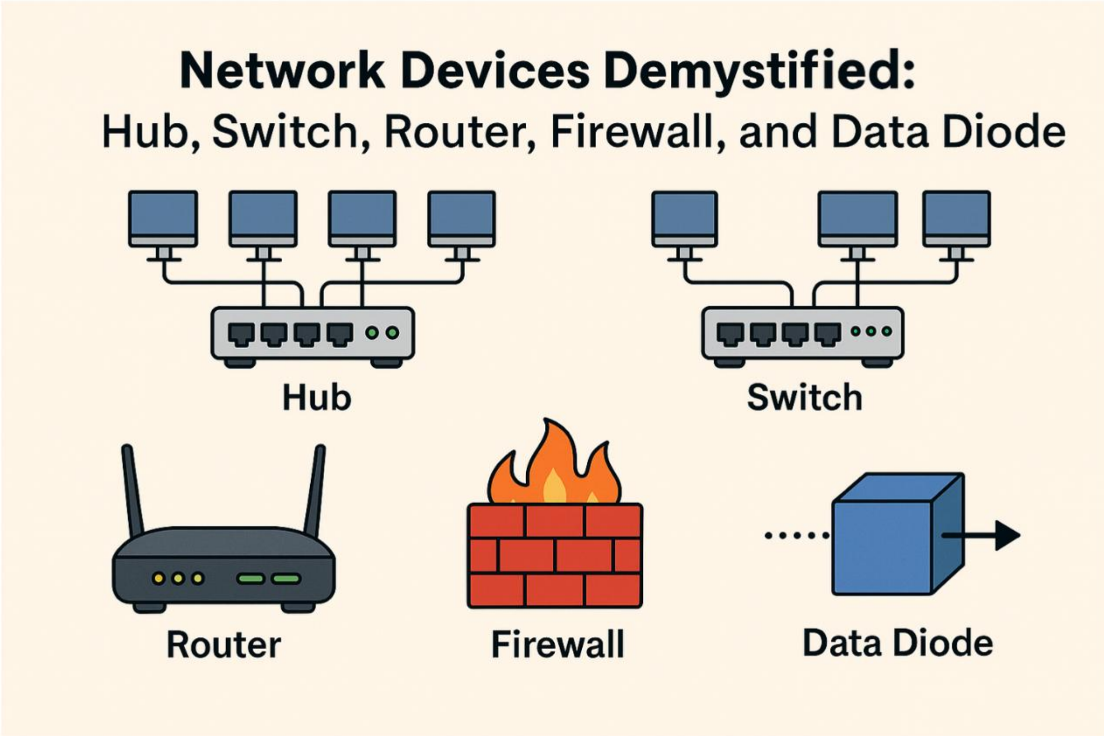

PC1
PC2
PC3
Every connection to the switch
is a separate collision
domain
Switch
Every device can
send and receive at
the same time

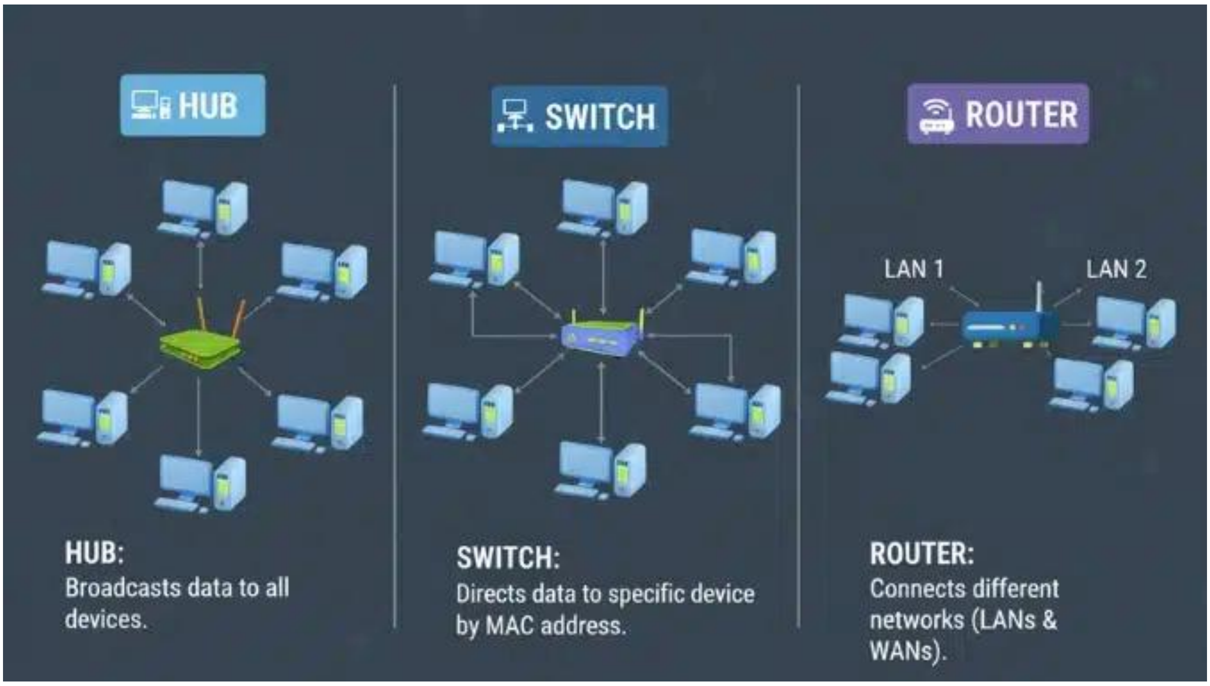

Network Switches:
Types & Components
What is a Network Switch?
Forwards data based on MAC addresses.
Types of Network Switches
Unmanaged Switch
Basic Plug & Play
Managed Switch
VLAN & Security
Layer 2 Switch
MAC Address Switching
Layer 3 Switch
Routing & IP Switching
PoE Switch
Power over Ethernet
Inside a Network Switch
Ethernet Ports
Switching Fabric
CPU / Processor
RAM / Memory
Power Supply (PSU)
Cooling Fan
LED Indicators



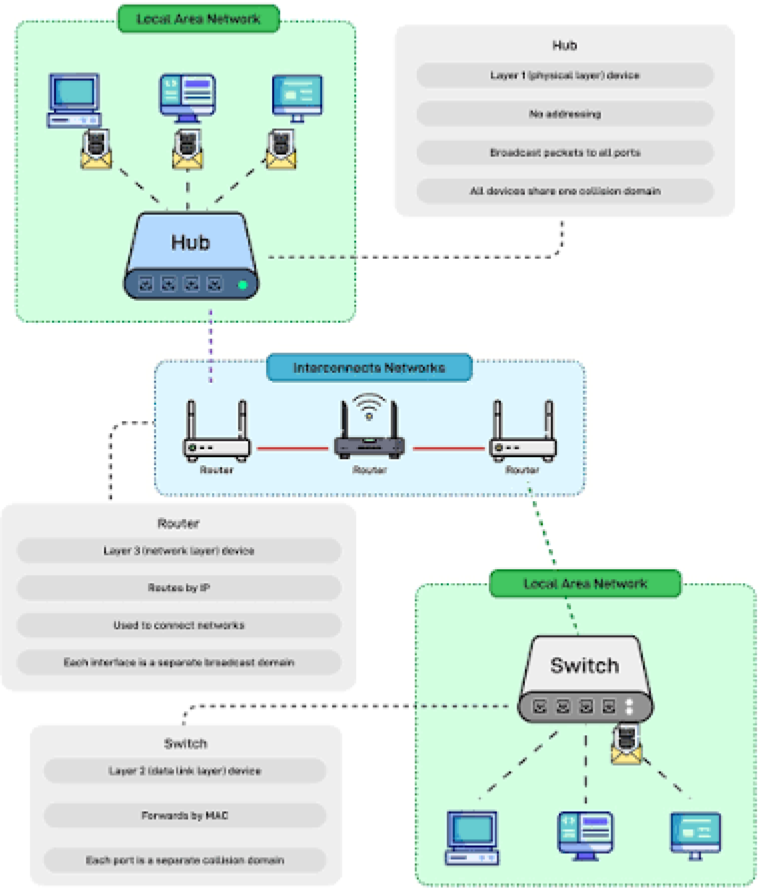
Hub vs. Switch vs. Router

Created by Dan Nanni study-notes.org	Hub	Switch	Router
Layer	Physical layer (L1)	Data link layer (L2)	Network layer (L3)
Function	Blindly broadcasts to every connected device	Unicast, multicast or broadcast selectively based on destination MAC address or VLAN header	Forward based on source/destination IP addresses (Other addon functions such as NAT/VPN/WiFi AP)
Data to Store	None	MAC table that maps MAC addresses to switch ports	IP routing table with route entries for each destination subnet
Traffic Change	None	None	Modify TTL & checksum in IPv4 header
Broadcast Domain	One broadcast domain	One broadcast domain per VLAN	Separate broadcast domains per interface
# Ports	4/12 ports	4-48 ports	2/4/5/8 ports
Delivery	Bits	Frames	Packets
Mode	Half duplex	Half/Full duplex	Full duplex


Network Devices Demystified:
Hub, Switch, Router, Firewall, and Data Diode
Hub
Switch
Router
Firewall
Data Diode

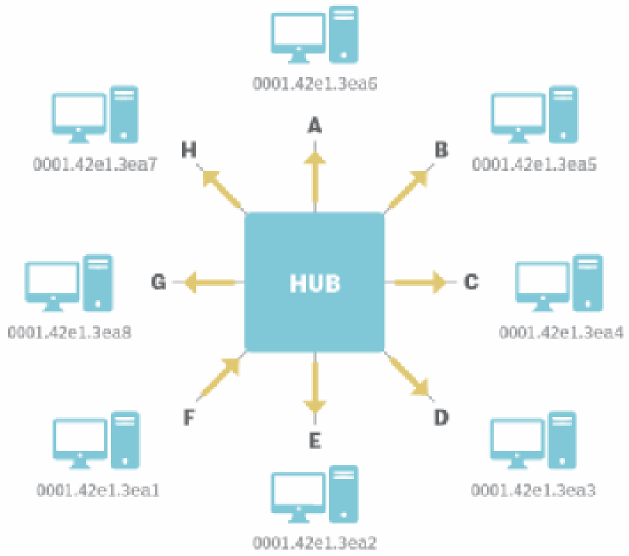

HUB
SWITCH
ROUTER
LAN 1
LAN 2
HUB:
Broadcasts data to all devices.
SWITCH:
Directs data to specific device by MAC address.
ROUTER:
Connects different networks (LANs & WANs).

Hub, Switch, & Router Explained

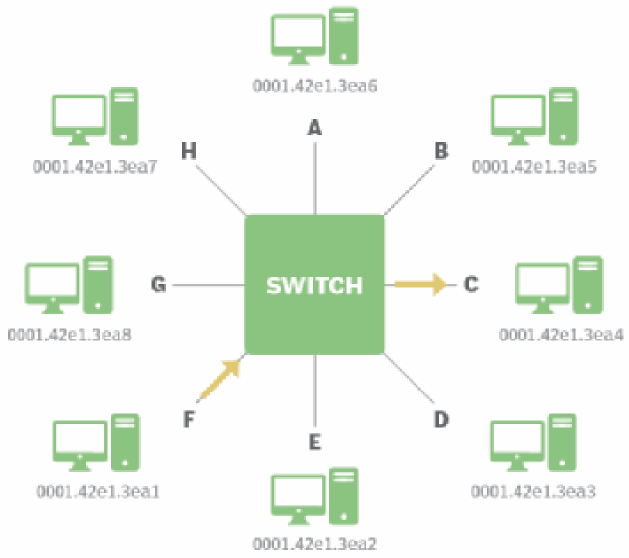



Switch vs Hub
Key Differences Explained
Network Switch
VS
Network Hub
Data Handling:
Sends data to specific device
Speed:
Faster Performance
Traffic Efficiency:
Reduces Network Congestion
Security:
Offers Better Security
Data Handling:
Broadcasts data to all ports
Speed:
Slower Performance
Traffic Efficiency:
Causes Network Collisions
Security:
Less Secure
Data Transfer
Data sent directly to the intended device
Data Transfer
Data sent to all connected devices
Key Points
Switch: Intelligent, sends data to the right device
Hub: Simple, sends data to all devices on the network
@all_computer_gyaan95
@computerhardwaresoftwarenetworking

Hub vs. switch

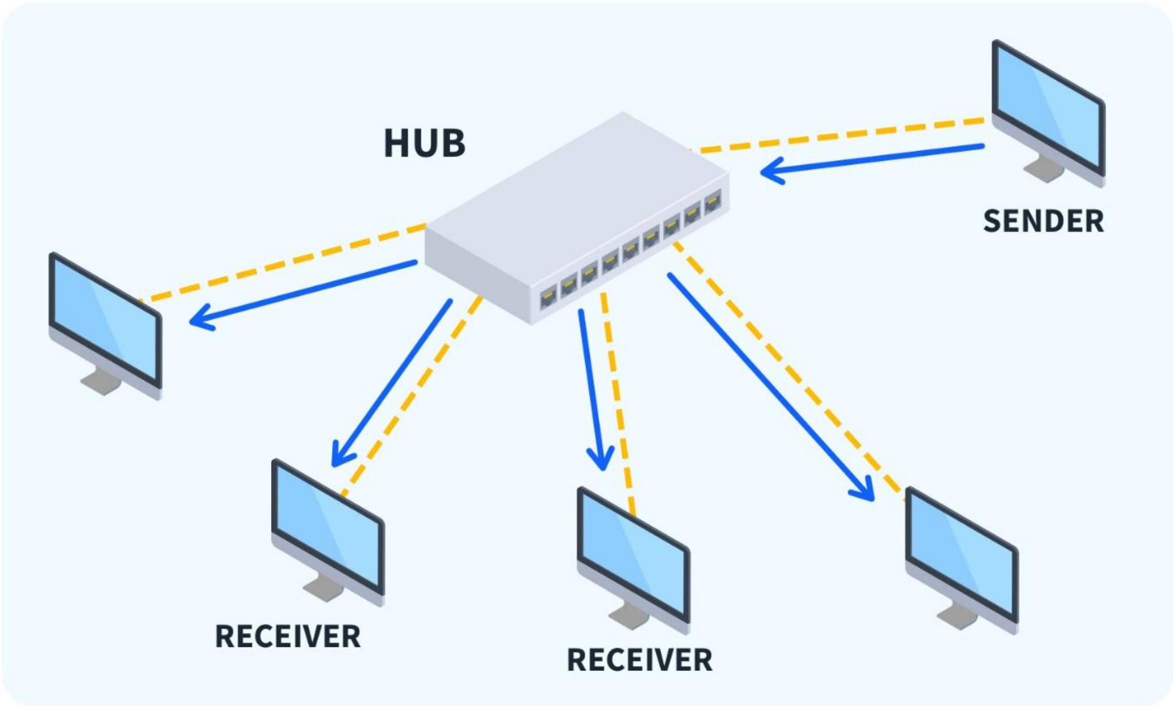


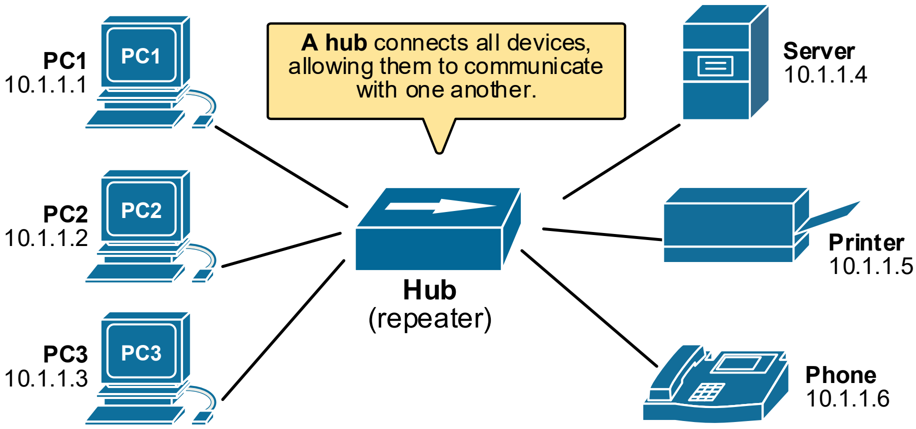
In this diagram of a **network hub**, the source, PC F - MAC 0001.42e1.3ea1, sends a packet to the destination, PC C - MAC 0001.42e1.3ea4. All other devices receive the communication, however. The message is broadcast to all other devices on the hub.



In this diagram of a **network switch,** the source, PC F - MAC 0001.42e1.3ea1, sends a packet to the destination, PC C - MAC 0001.42e1.3ea4. Only PC C receives the message because the switch has a table entry showing which port PC C is connected.

SOURCE: THE McGRAW-HILL COMPANIES, INC.


HUB
SENDER
RECEIVER
RECEIVER


A hub connects all devices, allowing them to communicate with one another.
PC1
10.1.1.1
PC2
10.1.1.2
PC3
10.1.1.3
Hub
(repeater)
Server
10.1.1.4
Printer
10.1.1.5
Phone
10.1.1.6